

Viewing tailored nudges is correlated with improved mastery-based assessment scores

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Abstract

Mastery-based assignments typically provide students with multiple opportunities to improve their performance, but getting students to take advantage of these opportunities is difficult. We report on the implementation of a two-part series of nudges designed to improve students' engagement with and performance on mastery-based assignments in high-enrollment, introductory undergraduate courses. Students received one or two email-based nudges for each assignment, providing both decision information and decision assistance. Delivered via email by a digital coaching system that provides tailored support to students, we evaluated the effectiveness of viewing versus not viewing these personalized nudges across introductory courses in mathematics, computer science and engineering over five terms at our large, research-intensive university. We used multilevel modelling with a sample of 30,693 assignment scores nested within 79 assignments and 5349 undergraduate students to explore the relationship between students' viewing the nudges and their scores while controlling for performance in other courses and students' baseline proclivities to read email from the coaching system. We find that viewing versus not viewing nudges is significantly related to improved outcomes. Viewing the first of the two-part nudge series is associated with a 3% increase in scores on average. Importantly, we found a significant interaction with prior academic performance

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such that students with the lowest average academic performance across all courses are predicted to benefit the most—up to 9%—from nudging. Positive impacts are observed in most courses where nudges were implemented and are found to be most impactful for later versus earlier assignments within a term. We discuss the implications of these results for supporting students' decision-making within mastery-based assignment schemes and suggest future research for tailored nudges to support student success.

KEYWORDS

alternative grading, higher education, mastery-based assessment, multilevel modelling, nudging, STEM, tailoring

Practitioner notes

What is already known about this topic

- Nudges are known to be broadly effective, but heterogeneity is observed especially in education where the research base is relatively limited.
- Mastery-based grading is an alternative approach to grading that, among other features, emphasizes learning through an explicit feedback cycle of struggle and revision without penalty.

What this paper adds

- This study assesses the effectiveness of personalized nudges in the context of mastery-based grading in undergraduate STEM courses.
- We examined the efficacy of viewing versus not viewing nudges in the real learning settings of three STEM courses enrolling more than 5000 students.
- We found the nudges to be effective overall, correlated with improvements of approximately 3% in students' assignment scores and most impactful for students with the lowest average academic performance.

Implications for practice and/or policy

- Nudging by email or text message has a low overhead cost and yet may provide materially significant gains in students' assessment scores.
- Nudges can be effectively targeted to specific subsets of students and tailored based on students' prior and concurrent learning analytics data.

INTRODUCTION

The competitive grading culture in large, introductory, undergraduate science, technology, engineering and mathematics (STEM) courses has been rampant for decades (Seymour & Hewitt, 1997). A recent keystone study on persistence and loss in undergraduate STEM degree programs shows that discouragement and lost confidence due to low grades in early courses are significant contributors to students leaving their STEM degree programs with the majority of students switching degrees citing it as a core concern (Seymour & Hunter, 2019).

In fact, the low grades in STEM relative to non-STEM courses (Tomkin & West, 2022) have been observed for more than 100 years (Meyer, 1908); the contributing practices of high-stakes exams and curved grading in early STEM courses are arguably part of the culture and habitus of these disciplines (Ballen et al., 2017; Bowen & Cooper, 2022).

Considering these challenges, interest in alternative forms of grading at the higher education level has been rising for at least 10 years (Clark & Talbert, 2023) in part based on a longer history of alternative grading success at the K-12 level (Iamarino, 2014; Marzano, 2011; Townsley & Varga, 2018). Examples of undergraduate courses using alternative grading systems—such as mastery-based grading and specifications grading—are now apparent across STEM disciplines (Beatty, 2013; Katzman et al., 2021; Toledo & Dubas, 2017; Zimmerman, 2020). According to Clark and Talbert (2023), the four core features of alternative grading systems are that learning objectives are clearly defined, feedback is helpful and generative, assignment grades indicate progress, and learning occurs through an explicit cycle of struggle and revision without penalty (Clark & Talbert, 2023). Alternative grading approaches are correlated with cognitive and affective gains for students (Beatty, 2013; Boesdorfer et al., 2018; Tarr et al., 2008) and evidence for concomitant gains for instructors is emerging (Lee et al., 2018; Prof, 2022), but the research base here is less robust.

Here, we focus on the core alternative grading feature of learning through struggle and revision. Within the implementation of mastery-based grading across three introductory courses, we investigate a tailored approach to supporting students in making multiple and earlier attempts at course assessments, providing both decision information and decision assistance to borrow from the language of choice architecture (Münscher et al., 2016). When students are offered the opportunity to improve over time in a lower-stakes flexible assessment system, they sometimes make choices to prioritize more rigid courses or other more timely demands and procrastination is a persistent concern (Rabin et al., 2011). We implemented a set of email nudge reminders using a web-based coaching platform, called ECoach. Prior research (Matz et al., 2021; McKay et al., 2012) has demonstrated the efficacy of ECoach features in general for improving student engagement and outcomes in STEM courses, however, the nudges studied herein are new to ECoach. In this paper, we investigate the notion that, within a flexibly designed assessment structure, viewing versus not viewing brief personalized nudges can encourage engagement and contribute to improved performance.

BACKGROUND

Thaler and Sunstein (2008) provided the canonical definition of nudges as ‘choice architecture that alters people’s behavior in a predictable way without forbidding any options or significantly changing their economic incentives’ (p. 6). Research on the effectiveness of nudges has largely taken place in economic and healthcare settings within the United States (Jia & Mustafa, 2023). Nudging in education has a shorter history; the first review on nudging in education (Damgaard & Nielsen, 2018) identified only 122 relevant studies of which nearly half had been published in the prior 2 years and Weijers et al. (2021) recently commented that ‘the popularization of nudging has not yet reached the field of education’ (p. 885). In higher education, extra credit is often used as a motivating ‘nudge’ for students (Padilla-Walker, 2006) but extra credit is an economic incentive and, as such, it falls outside the canonical definition of nudges (Selinger & Whyte, 2012). Even given the relatively limited research on nudges in education, a variety of nudge types have been used, from decision assistance and decision information to social group comparisons and changing defaults. Damgaard and Nielsen (2018) conclude that across the different types of nudges,

the greatest impacts generally occur 'for individuals affected most by the behavioural barrier targeted by the intervention' (p. 313).

Positive results from nudging are evident in a variety of domains and for different educational outcomes (Motz et al., 2021; Rodríguez et al., 2022). In the undergraduate context, for example, Smith et al. (2018) showed that nudging improved homework scores in introductory economics courses. Dart and Spratt (2021) reported improved academic performance for students in one of two introductory mathematics courses, like the context herein. Specifically in mastery-based courses, we find evidence that nudges can improve time-on-task (Eyink et al., 2020) and learning performance for middle school mathematics students (Vanacore et al., 2023). On balance, nudging in educational settings appears to have positive effects (as a case in point, of the 10 studies included in Lim and Lee's (2022) recent review, nine reported positive effects), yet some heterogeneity is still evident. A recent study in graduate public health education, for example, showed that nudging students to complete coursework in order to improve their grade outcomes had no effect (Garbers et al., 2023). Dobronyi et al. (2019) similarly found no effect of goal-setting interventions and follow-up reminders in a large sample of Canadian undergraduate students and surmised that the null result might be due to the sample being unselected for any particular characteristics (eg, low motivation).

Several explanations are available as to why nudges are sometimes ineffective. Sunstein (2017) has argued broadly that one reason nudges fail is that those being nudged might have very strong preferences and Entwistle (2021) has claimed that nudges fail for people with fatalist mindsets. The characteristics of nudges largely developed in economic contexts may not transfer particularly well to educational contexts; Weijers et al. (2021) claim that the long-term nature of educational goals and the ages of those targeted for nudging in education are substantially different. Damgaard and Nielsen (2018) call for 'better targeted nudges' (p. 339) or nudges that are more personalized to each individual and Mills (2022) similarly proposes that greater personalization will improve efficacy. With evidence that broadly applied homogeneous rules can have unintended effects and even harm specific subpopulations (Thunström et al., 2018), understanding how to personalize nudges in learning environments is a valuable direction for current research. Research on nudges needs to move past their efficacy and towards an understanding of mechanism and generalization (Kizilcec et al., 2020): how do nudges work for specific individuals in specific settings?

The virtue of the current study on personalized nudging in the context of mastery-based grading in higher education is its design and sample, which is balanced in its ability to understand both mechanism and generalization. First, the correlation between nudging and assignment scores is measured within-student and -course. That is, students chose whether or not to view each nudge, making for a clean comparison (nudged vs. not) that cannot be explained by differences between students or courses though we note these conditions were not randomly assigned. There was no counterbalancing of repeated factors, meaning that we cannot claim that the nudging caused any difference given the remaining dependencies between students and factors such as the ordering and frequency of nudge conditions, however, a within-subject design is free of major between-unit confounds. Second, our sample is sizable and diverse which allows us to examine—with adequate statistical power—for whom and under what situations nudges work.

CONTEXT

ECoach at the University of Michigan

With approval from the institutional ethics board, this study was conducted at the University of Michigan (U-M), a large, public, university in the United States with very high research

activity and predominately residential students (American Council on Education, [n.d.](#)). Nudges were implemented using a software tool called ECoach. The development and implementation of ECoach began over a decade ago in the context of introductory physics courses (McKay et al., [2012](#)) with the goal of giving tailored advice to students. ECoach was scaled over several years for use across introductory STEM lecture courses where student-to-faculty ratios on the order of several hundred make direct communication between instructors and individual students difficult.

The core student-facing features of ECoach are messages about key course elements, metacognitive interventions to support studying for and reflecting on high-stakes exams and an interface for exploring possible grade outcomes. Tailoring with personalized content across these features is accomplished based on information that students provide on a substantive survey. Survey topics include, for example, prior experiences with the course subject, dispositions around high-stakes exams and level of motivation. The tailoring engine is also connected to the learning management system and the student data warehouse. ECoach now supports more than 20,000 undergraduate student enrollments annually across nine large U-M STEM courses as well as students in similar courses at a few partnering universities. Research has demonstrated the efficacy of engagement with ECoach in general for improving students' course grades (Huberth et al., [2015](#); Matz et al., [2021](#)) and the particular effectiveness of the main metacognitive intervention for exam study (Chen et al., [2017](#), [2022](#)).

While influencing some student behaviours (like how to study for an exam) requires in-depth advice, other behaviours (like starting an assignment earlier than usual) can be better influenced by simple reminders. We have used ECoach to accomplish the latter in two ways to date. First, Brown et al. ([2019](#)) studied the effects of two different social comparison nudges on students' starting their homework earlier in a large introductory statistics course. While there was no overall main effect, the authors observed an interaction in which nudged students who felt earning their goal grade was less important showed earlier homework start times compared to students in the control condition with the same goal grade importance who did not receive a nudge. Second, as preliminary work underlying the current study, we assessed the correlation between viewing versus not viewing behavioural nudges and students' mastery assessment scores in one term of an introductory mathematics course, finding improvements of 0.3–0.9 points on a 5-point assessment scale (Matz et al., [2022](#)).

Courses and nudge design

The nudges in this study were implemented across three introductory STEM courses, herein called Math 101, Computer Science 101 and Engineering 101. The vast majority of students in these courses are enrolled because the courses are part of a required sequence within their degree program.

Math 101 is a precalculus course enrolling 500–600 students per year in small sections of about 20 students each. The course faculty recently overhauled the course assessment scheme, whereas grades used to be essentially based on three high-stake exams, the new scheme includes mastery-based assessments in addition to lower-stake exams and regular homework. The mastery assessments typically include five questions on a given topic. Students have about 10 days to complete the assessment in a proctored setting and can retake it up to twice per day during that timeframe to improve their score (credit is given only above a certain threshold). In the initial shift to mastery assessments, however, students frequently waited to make their first attempt and used few available attempts overall.

To help students take advantage of the multiple retake design, a series of two email nudges (Figure 1) was implemented using ECoach. The goals of the nudges were to encourage students to make their first attempt earlier, to provide students with feedback about their current score and their number of remaining attempts and to adjust norms so that students would see initially low scores as okay places to start. The first nudge was sent to all students 5 days ahead of the due date with their current score, their number of remaining attempts and tailored feedback. The second nudge was sent 2 days ahead only to students who had not yet earned a passing score and was intentionally designed to be more succinct as the deadline was closer. The second nudge again included their current score, their number of attempts remaining and brief tailored feedback. For example, this nudge was different for students who had not yet started the assignment versus students who had made multiple attempts but had not yet passed.

Computer Science 101 and Engineering 101 are programming courses, the former for a subset of computer science majors and the latter for all engineering majors. Both include coding projects in the assessment scheme. The projects are submitted to an automatic grading system that evaluates students' code and students could submit their projects up to four or five times daily. Thus, a nudging scheme modelled after that for Math 101 was implemented in these courses with similar goals of helping students start early and keep track of their current score and number of attempts remaining.

Hi, Victor. Here's a summary of where things stand on your Mastery Assessments this week: Mastery09_Proctored - Score: 3 out of 5 (only scores of 4 or more get credit) - Attempts remaining: 10 Keep going! It looks like you've been trying hard — that's great. Use the feedback you get to figure out what to study more. If you're getting the same ones wrong each time, bring them to office hours.	Hi, Victor. Looks like you don't yet have a passing score on Mastery09_Proctored, but you can still get there. What's one thing you can practice so you can try again? Is that something you can do today? How can you use your remaining 3 attempts before Thursday? Remember: you only get credit for scores of 4 or 5.
Hi, Michael. Here's a summary of where things stand on Project 5. Project 5 details - Highest score so far: You haven't started yet - Submits remaining: 20 - Overall project average so far: 98.5 Start with one part. Big projects can be daunting, so maybe try to break it down. Make a plan in pseudocode first. From there, you can code one part and submit it to see what to improve. One step at a time is far less overwhelming. Tip of the week You want to prove to yourself you can do well in this course. We love that kind of energy! Here's the good news: [course] is set up to reward people who try really hard. So it's more about the effort you put into it, and less about what you know on Day 1.	Hi, Michael. Just a reminder: you haven't submitted Project 5 to the Autograder yet, but there's still time. You still have 4 submits. Remember to do some testing on your own as well. Remember: You can post questions (but not code) on Piazza or visit office hours for help.

FIGURE 1 Examples of nudge 1 (left column) and nudge 2 (right column) for Math 101 (top row) and Computer Science 101 and Engineering 101 (bottom row; the project-oriented nudges for these two courses were similar). Additional examples are provided in the [supplementary material](#).

METHODS

Participants

Students were nested across 11 ECoach 'coaches', where a coach indicates a unique pairing between an academic term and the course of interest between Fall 2020 and Fall 2022 (Table 1). The sample includes 5349 enrollments from 5211 unique students as some students were enrolled in more than one coach, however, we refer to students and not enrollments throughout for simplicity. Students had between 4 and 10 assignments over the term depending on the coach. The majority (73%) of students were White or Asian with a balanced split on sex (54% male). Students who withdrew were excluded.

Measures

Assignment score

The outcome measure was the student's final assignment score. As assignments differed in point value, scores were standardized to a common scale by dividing each student's score by the maximum for each assignment. Thus, scores are measured on a 0 to 1 scale. Scores of 0 represent both students who did not turn in an assignment and those who attempted an assignment and earned a score of 0. The average score by coach is shown in Table 1.

Primary and secondary nudge email engagement

We examine the extent to which viewing versus not viewing the nudge emails is related to final assignment scores and note that we cannot parse viewing nudges from reading nudges. That is, ECoach provides the timestamp indicating when the nudge was viewed (opened) and no means to determine how long students engaged with it, if at all. The timestamp is generated based on a tracking pixel embedded in each email. While such data can be limited by particular privacy settings or browser extensions, the default settings of the email client supported at U-M support tracking pixels. The average proportion of students who viewed each nudge by coach is shown in Table 1.

Non-nudge email engagement

Students also received between 9 and 36 non-nudge-related emails from ECoach depending on the course and term (greater values result from students who opted into a set of 12 'bonus' emails about topics like multitasking). To control for differences between students in their proclivity to engage with ECoach email in general, we computed each student's fraction of non-nudge emails viewed. Email engagement is assumed to reflect a student's baseline level of engagement with ECoach with greater email viewership indicating more engagement. We use this value as a control to probe the effect of nudge emails on assignment scores at fixed engagement. That is, the relationship between viewing versus not viewing nudges and assignment score is calculated at a given level of platform engagement, which controls for differences between students in platform engagement. The average proportion of other emails viewed by students by coach is shown in Table 1.

TABLE 1 Characteristics of the student population within each coach.

Course	Term	Assignments (N)	Students (N)	Average nudge 2 emails (N)	Assignment score		Proportion of eligible students who viewed		Proportion of other emails viewed		GPAO	
					M	SD	Nudge 1 (M)	Nudge 2 (M)	M	SD	M	SD
MTH	F20	10	419	257	0.88	0.22	0.67	0.67	0.66	0.25	3.59	0.51
MTH	W21	9	91	71	0.87	0.22	0.59	0.56	0.69	0.28	3.57	0.37
MTH	F21	10	507	322	0.88	0.21	0.80	0.79	0.80	0.23	3.45	0.54
MTH	W22	10	129	87	0.91	0.16	0.79	0.79	0.71	0.24	3.41	0.47
MTH	F22	6	481	257	0.86	0.20	0.88	0.73	0.78	0.20	3.52	0.49
CS	F21	6	628	190	0.74	0.34	0.80	0.78	0.81	0.23	3.69	0.36
CS	W22	6	507	181	0.74	0.36	0.83	0.82	0.75	0.22	3.66	0.37
CS	F22	6	1175	850	0.97	0.09	0.68	0.81	0.75	0.21	3.61	0.48
ENG	F21	6	431	130	0.86	0.28	0.77	0.73	0.74	0.29	3.49	0.53
ENG	W22	6	346	127	0.87	0.28	0.77	0.75	0.70	0.25	3.40	0.48
ENG	F22	4	635	318	0.93	0.22	0.57	0.53	0.68	0.22	3.52	0.47

Note: The courses are Math (MTH), Computer Science (CS) and Engineering (ENG). The terms are fall (F) and winter (W). The number of nudge 1 emails sent per assignment is equal to the number of students in the coach. Because the number of nudge 2 emails sent per assignment is variable, we report the average number sent across assignments. CS 101 students could earn bonus points on their assignments for turning in projects early but these points are excluded here.

Grade point average in other courses

Grade point average in other courses (GPAO) is a student's grade point average in all courses overall terms other than the course in which the nudge effect was measured (Huberth et al., 2015; Young et al., 2023). GPAO is a general academic performance metric that reflects a student's baseline level of college ability and has greater predictive power than standardized exam scores or high school GPA (Koester et al., 2016). We use GPAO as a control in examining the relationship between viewing the nudges and assignment scores. That is, the effect of the nudges is the expected assignment grade increase at a given GPAO, which controls for differences between students in college ability. The average GPAO for students by coach is shown in Table 1.

Analytic method

The extent to which the primary and secondary nudges (specified as 0=not viewed, 1=viewed), non-nudge email engagement and GPAO were related to assignment scores was examined in a sample of 30,693 scores nested within 79 assignments and 5349 students. Non-nudge email engagement and GPAO were centred at 50% and 3.0, respectively, one standard deviation below the sample means of 74% and 3.5, respectively, because values of zero are not particularly meaningful for these variables. We centred the variables in this way because we expected the nudges to be more impactful for students in the bottom (rather than the top) halves of these distributions. The difference ($N=4624$) between the expected number of scores across all coaches ($N=35,317$ based on Table 1) and the modelled sample ($N=30,693$) arises from a few unique scenarios. For example, one mathematics assignment was excluded because students were sent an additional set of nudges after the deadline was extended following a graduate student assistant strike. As another example, at least one assignment from each course was excluded in Fall 2022 due to a short-term database error that resulted in email view events going unrecorded.

Assignments and students were crossed and nested within the 11 coaches given that each student experienced each assignment within a single coach (Figure 2). Multilevel modelling in SAS (code available upon request) accommodated this cross-classified sampling structure. The model included a student-level random intercept to control for mean differences in scores between students and a student-level random nudge slope to control for mean differences in the effect of the nudges between students. The relatively small number

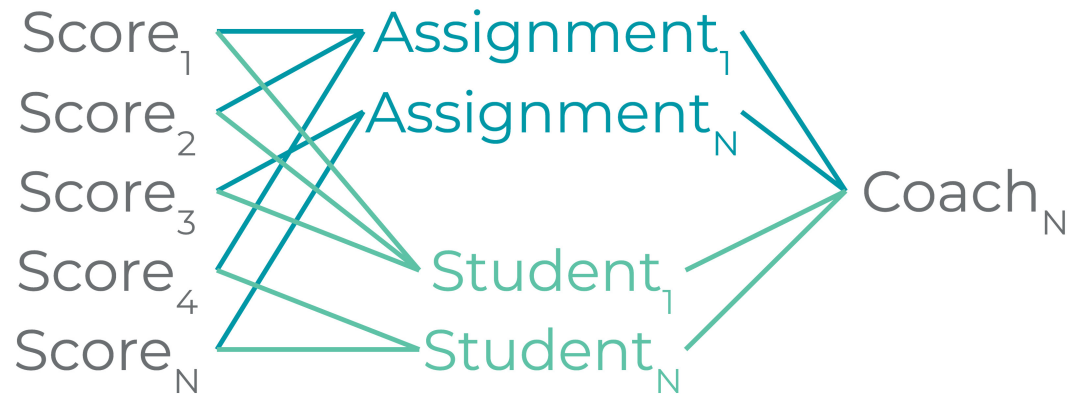


FIGURE 2 Diagram showing the crossed and nested nature of scores, assignments and students within coaches.

of coaches and assignments within a coach precluded the estimation of random effects for these dimensions. As such, these terms were included as fixed effects to control for mean differences in scores between coaches and between assignments (Hoffman, 2015). Additionally, the coach variable was allowed to interact with the nudge, non-nudge email engagement and GPAO variables, which controls for differences between coaches in these effects. The assignment variable was allowed to interact only with the effect of the nudges (non-nudge email engagement and GPAO were student-level factors and therefore did not vary across assignments), which controls for differences between assignments in the effect of nudge. The model is effectively a cross-classified model with random intercepts and nudge slopes for each sampling dimension—students, assignments and coaches.

RESULTS

Table 2 shows the model estimates for the relationships between viewing versus not viewing nudge 1 (aggregating over all nudge 2 conditions), GPAO and non-nudge ECoach email engagement with assignment scores for the average coach and assignment. The intercept indicates that a student who is one standard deviation below average in non-nudge email engagement (50%) and GPAO (3.0) in an average coach earned 78% on an average assignment for which nudge 1 was not viewed. Importantly, there was a significant relationship between viewing nudge 1 and scores for these students such that those who viewed the nudge scored 3% higher than those who did not, on average.

Role of student GPAO and email engagement

As expected, assignment scores were significantly higher for students with higher GPAOs. Scores also tended to be higher for students with higher email engagement, though this effect was not significant. This result is likely due to the covariance between email engagement and GPAO such that after controlling for GPAO, there remains no unique relationship between email engagement and score. In other words, the extent to which students engage with email provides no predictive utility of scores beyond that provided by student GPAO.

Importantly, the correlation between viewing nudge 1 and assignment scores interacted significantly with GPAO for the average coach and assignment implying that the nudges had a larger impact on students with lower GPAOs (Figure 3, left panel). Similarly, the nudges had a numerically larger correlation for students with lower email engagement,

TABLE 2 Model estimates, standard errors of the mean (SE) and *p*-values for the relationship between viewing nudge 1, non-nudge ECoach email engagement, and GPAO with assignment score for the average coach and assignment.

Parameter	Estimate	SE	<i>p</i> -value
Intercept	0.7790	0.0071	<0.0001
Nudge 1 (0 = not viewed)	0.0298	0.0068	<0.0001
Email engagement (0 = 0.50)	0.0380	0.0256	0.1368
GPAO (0 = 3.0)	0.0949	0.0099	<0.0001
Nudge 1 × email engagement	−0.0417	0.0248	0.0919
Nudge 1 × GPAO	−0.0191	0.0094	0.0422
Email engagement × GPAO	−0.0245	0.0363	0.5000
Nudge 1 × email engagement × GPAO	0.0409	0.0350	0.2424

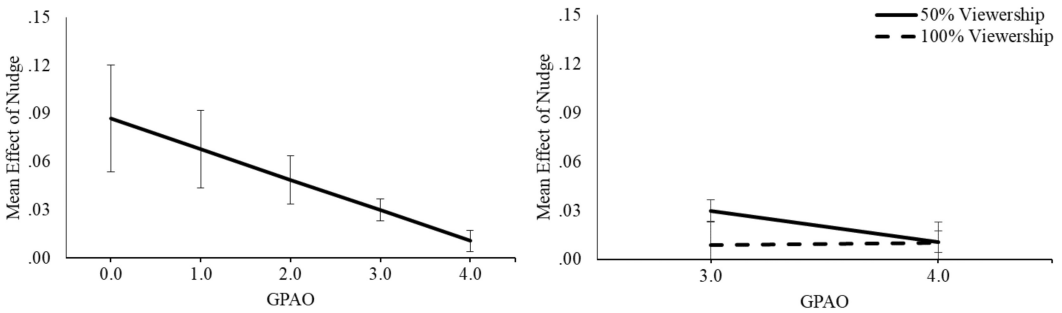


FIGURE 3 Mean effect ± 1 standard error of nudge 1 as a function of GPAO (left panel) and specifically for students with 50% and 100% non-nudge ECoach email engagement (right panel).

though the interaction of nudges and email engagement was not significant. The interaction of nudge 1, GPAO, and email engagement also was not significant. Numerically, the interaction pattern was such that the larger nudge-related score increase at lower GPAOs weakened with more email engagement (Figure 3, right panel). Again, the lack of significant interaction with email engagement was likely due to its covariance with GPAO. Taken together, we conclude that, for an average coach and assignment, the relationship between viewing nudge 1 and improved assignment scores is especially strong for those who need it most (ie, those with lower GPAOs), regardless of their level of general email engagement.

Differences between coaches

The apparent effectiveness of nudge 1 differed significantly between coaches, $F(10, 2488) = 2.97, p = 0.001$. Figure 4 shows the nudge 1 effect by coach for students one standard deviation below the mean in non-nudge ECoach email engagement (50%) and GPAO (3.0); there are clear differences between coaches. Error bars that do not overlap 0% indicate significant effects ($p < 0.05$). Seven of the 11 coaches showed a significant positive relationship between nudge 1 and assignment scores. A significant effect was observed at least twice (ie, in two different terms) for each course demonstrating some degree of generalization across disciplines, courses, terms and instructors. Interestingly, three of the four failures to replicate occurred in Math 101; the different student populations in Computer Science 101 and Engineering 101 versus Math 101 (which is specifically a preparatory course) or the differences between the mastery-based programming projects and math assignments could be driving this interaction of nudge and coach.

Differences between ordinal assignment number

The apparent effectiveness of nudge 1 differed significantly between assignments, $F(9, 21,239) = 2.41, p = 0.009$. Figure 5 shows the predicted mean assignment score by ordinal assignment number for students who did and did not view the email nudge (specifically for those with GPAO = 3.0 and email engagement = 50%). Because assignments in these courses are distributed evenly throughout the term, ordinal assignment number can essentially be thought of in terms of time; that is, overall, the relationship between viewing nudge 1 and improved assignment scores appears to have become more effective over the term. One possibility is that nudges become more effective as the term gets more difficult,

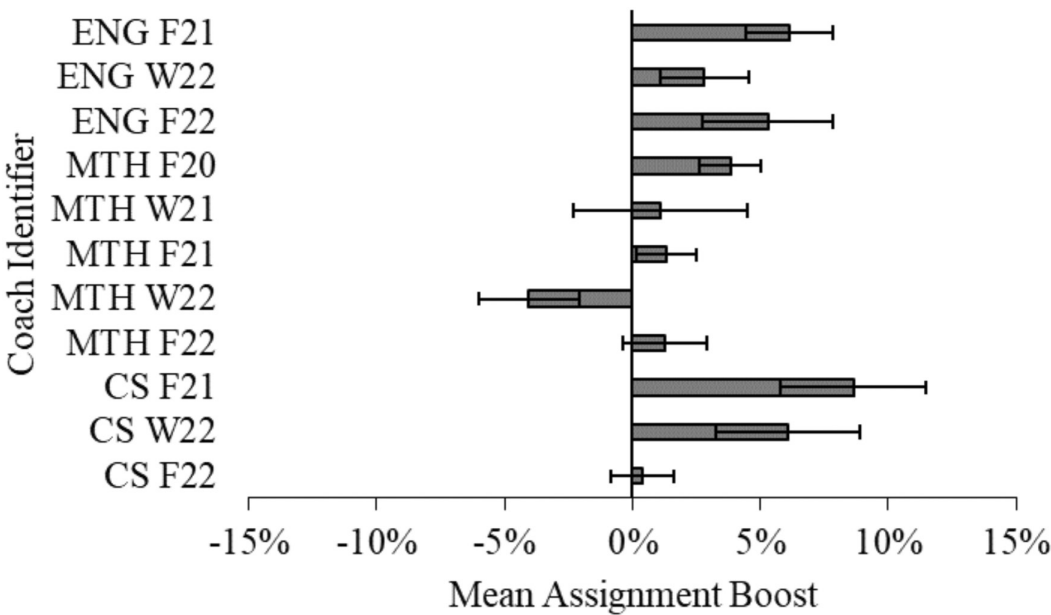


FIGURE 4 Mean assignment score change ± 1 standard error for viewing nudge 1 in each coach, specifically for students with GPAO=3.0 and non-nudge ECoach email engagement=50%.

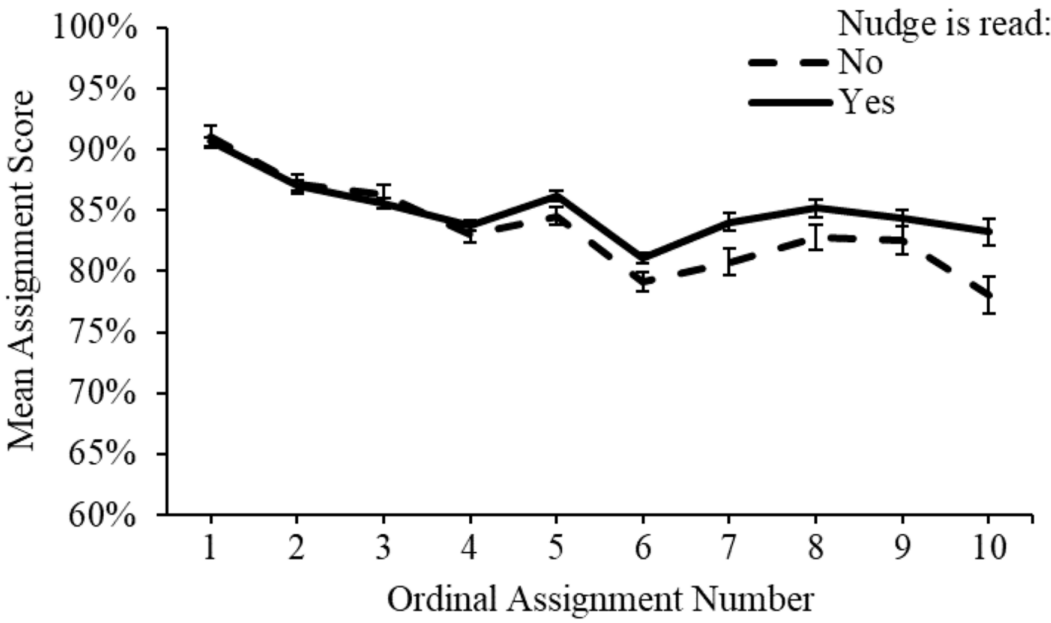


FIGURE 5 Predicted mean assignment score ± 1 standard error by ordinal assignment number for students who did and did not view nudge 1.

perhaps due to elements of the course itself or the broader context of the term (eg, other courses, work, family obligations, etc.).

To test the functional form of this difference, we predicted assignment scores with the same model as above except that the ordinal assignment number was specified as a quadratic function with random linear and quadratic slopes for students and a

fixed interaction with the nudge. The model was supported by a significant interaction of nudge 1 and quadratic assignment number ($Est=0.0012$, $SE=0.0006$, $p=0.0340$) such that the decrease in scores over the term lessened for students who viewed the nudge ($Est=0.0018$, $SE=0.0003$, $p<0.0001$) but not for those who did not ($Est=0.0006$, $SE=0.0005$, $p=0.2350$). Figure 6 (left panel) shows this difference in quadratic change. The net result was a stronger relationship between nudge 1 and assignment score over the term, the form of which followed a quadratic function (Figure 6, right panel). In sum, there was a nonlinear change in the apparent effect of nudge 1 such that the nudge-related increase in scores accelerated over the term. It is possible that this change was commensurate with an accelerating difficulty of the assignments, the course or the term more generally such that nudges were more effective the more difficult the local or global context of the student's learning environment.

Nudge 2 considerations

All students received a first nudge reminding them to complete their assignment with feedback on their current score, if any. A second nudge followed if a student had not yet completed their assignment with a certain threshold score. Table 3 shows the possible relationships between nudge 1 and nudge 2. To this point, we have examined differences in assignment scores between those who did and did not view the first nudge. In other words, we compared nudge conditions 1, 2 and 3 (in which nudge 1 was not viewed) with conditions 4, 5 and 6 (in which nudge 1 was viewed), which represents the relationship between

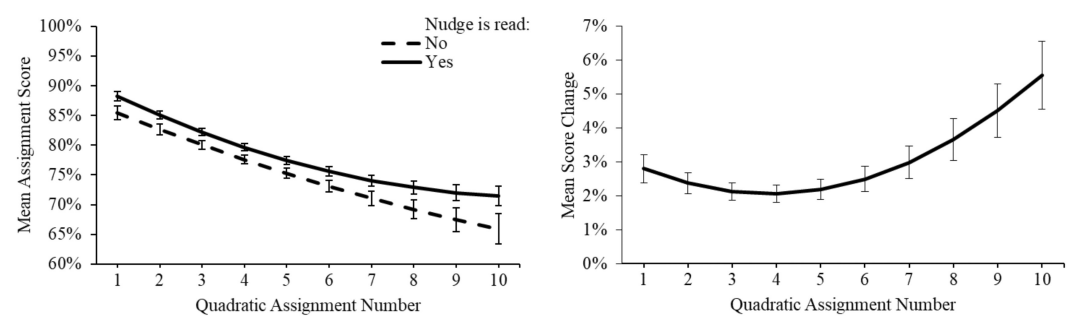


FIGURE 6 Predicted mean assignment score (left) ± 1 standard error as a function of quadratic assignment number for students who did and did not view nudge 1, and score change (right) ± 1 standard error as a function of quadratic assignment number.

TABLE 3 Possible conditions (Cx) for a given student and assignment considering the possible relationships between nudge 1 and nudge 2.

Cx	Nudge 1	Nudge 2	Assignment scores	
			N	M
1	Not viewed	Not viewed	2682	0.844
2	Not viewed	Viewed	2717	0.875
3	Not viewed	Not applicable	2319	0.939
4	Viewed	Not viewed	2085	0.839
5	Viewed	Viewed	11,144	0.860
6	Viewed	Not applicable	8368	0.944

viewing nudge 1 and assignment scores aggregating over nudge 2 conditions. In situations, where nudge 2 was both sent and viewed (conditions 2 and 5), this means that the effect of nudge 1 also included some portion of the effect of nudge 2. As such, we now disentangle the effects of nudges 1 and 2.

To parse the effects of nudges 1 and 2, we included predictors for each. One predictor was for nudge 1 (0 = not viewed) and the other was a set of dummy codes for nudge 2 (0 = not viewed). Both were allowed to vary over students, assignments and coaches. Thus, other than the addition of fixed and random effects for nudge 2, the model was identical to that described previously. We found a significant relationship between viewing nudge 1 and assignment scores for each condition of nudge 2 such that viewing nudge 1 was associated with higher scores, on average, regardless of whether nudge 2 was not viewed, viewed or not applicable (Table 4); this positive increase for assignment score ranged from 4% to 17%. In particular, the effect of nudge 1 was significantly larger by a factor of four when nudge 2 was not applicable relative to when nudge 2 was not viewed (Est = 0.1333, SE = 0.0129, $p < 0.0001$) or viewed (Est = 0.1317, SE = 0.0106, $p < 0.0001$). In other words, nudge 1 was apparently more effective for students who did not qualify for nudge 2 than for students who did.

There was also a significant relationship between viewing nudge 2 and assignment scores when nudge 1 was not viewed such that viewing nudge 2 was associated with 3% higher scores, on average. In contrast, there was no effect of nudge 2 when nudge 1 was viewed, indicating no cumulative benefit for additional nudges on the same assignment. Finally, we examined magnitude differences in the effects of nudge 1 and nudge 2 by comparing the effect of nudge 1 when nudge 2 was not viewed with the effect of nudge 2 when nudge 1 was not viewed. We found no significant difference (Est = 0.0066, SE = 0.0137, $p = 0.6294$), indicating that each was equally beneficial when the other was not viewed.

DISCUSSION

We find that viewing versus not viewing email nudges is generally correlated with improved mastery-based assignment scores and that the improvements are larger for students with lower versus higher grades in their other courses. We observe improved scores in relation to the nudges in each of the introductory courses included here. We find that nudges are apparently more effective for later versus earlier course assignments. In deconfounding the effects of nudge 1 and 2, we find that the first nudge is effective regardless of the nudge 2 condition (ie, whether nudge 2 is viewed, not viewed or not applicable; of these, nudge 1 is

TABLE 4 Model estimates, standard errors of the mean (SE) and p -values for the effects of nudges 1 and 2 on assignment score for the average coach and assignment and for their difference when the other was not viewed.

Term	Estimate	SE	p -value
Nudge 1: nudge 2 not viewed	0.0388	0.0144	0.0071
Nudge 1: nudge 2 viewed	0.0403	0.0134	0.0027
Nudge 1: nudge 2 not applicable	0.1721	0.0142	0.0001
Nudge 2: nudge 1 not viewed	0.0322	0.0149	0.0314
Nudge 2: nudge 1 viewed	0.0015	0.0119	0.8975
Difference: nudge 1–nudge 2	0.0066	0.0137	0.6294

most effective when nudge 2 is not applicable). The second nudge is effective only if the first nudge goes unviewed, meaning there is no dosage or cumulative effect in this context, a result that is possibly explained by nudge 2 being shorter than nudge 1. The magnitudes of the effect of viewing one nudge when the other is not viewed are equivalent. Taken together, we recommend nudging students until they view the nudge, at which point no additional nudges are necessary.

The effect size of the nudges is materially significant. The low overhead cost of viewing an email in relation to a 3% average improvement on a single assignment should make nudges like these attractive to instructors in both alternative grading and traditional grading scenarios, especially for two reasons. First, assignment scores here were relatively high (averaging 74% to 97%), in which case we would expect the effect of nudges to be blunted as scores approach the ceiling. Second, students were relatively high-performing (mean GPA = 3.5) and academically engaged (mean ECoach email open percentage = 75%) in which case we also would expect the effect of nudges to be blunted as students approach ceiling performance. In these less-than-ideal conditions where ceiling performance is expected, we still found reliable effects. The results appear to be more consequential for students with below-average performance and engagement.

The findings corroborate other efficacy studies of personalized nudges in educational contexts. In particular, an earlier study of social comparison nudges implemented via ECoach showed an interaction where students who felt earning their goal grade was less important showed earlier homework start times when nudged than control students (Brown et al., 2019). This result is like the finding observed herein where students with lower GPAOs were predicted to show a stronger relationship between nudge viewing and assignment scores compared to students with higher GPAOs. Both findings confirm the conclusion of Damgaard and Nielsen (2018) that those exhibiting behavioural barriers to the outcome at hand are most likely to show a positive effect from nudging. Zavaleta-Bernuy et al. (2022) also showed that nudge emails, though personalized to a somewhat lesser extent than those here, yield modest results with similar magnitudes of improved homework scores, but no clear impact on how early students begin their homework. In contrast, text messages reminding chemistry students to study had similar effects that mitigated exam grade differences between 'at risk' and other students though they were not personalized (Ye et al., 2016), and Plak et al. (2023) found tailored nudges to be no more effective than untailored nudges.

Limitations

Each assignment in this study was open for a few days before the first nudge was sent. In this timeframe, students could have submitted their assignment and earned full credit, rendering the first nudge unnecessary. However, the timestamp at which the assignment was submitted is unavailable in our data, thus we cannot parse these students from the sample. The effect of these students being in the sample acts in opposing directions. The students who completed the assignment with full credit yet still viewed the first nudge email effectively inflated the results. The students who completed the assignment with full credit and did not view the first nudge effectively deflated the results. Because these effects are in opposite directions, and the number of students who submit assignments early is likely to have been small (Felker & Chen, 2022; Nieberding & Heckler, 2021), we contend that this limitation in the data is unlikely to unduly affect the results. Additionally, we remind the reader that the results of this quasi-experimental study do not necessarily imply causation; additional randomized trials would be necessary towards this end.

Implications

The key implication of this study is that students appear to only need nudging once per assignment. Students who view a first nudge need not be sent a second and conversely, if students do not view a first nudge, some benefit is still expected from them viewing a second. The former case is important to implement as students report being annoyed by multiple reminders if they are already working on an assignment or have planned time to complete it (Zavaleta-Bernuy et al., 2022). Further, if resources are limited, nudges should be prioritized for later versus earlier assignments.

Future work

Other tailoring elements, particularly the means, target, goal and strategy of nudges (Bimba et al., 2017) are worth investigating and nudging differentially based on assignment difficulty or on how students were nudged and subsequently performed on a prior assessment would be illuminating. There is also a continued need to explore the cognitive processes and behavioural mechanisms at play when students are nudged (Damgaard & Nielsen, 2018; Weijers et al., 2021). Indeed, seeing a targeted email subject line itself, as opposed to opening a nudge email, may provide an adequate attentional cue to achieve similar effects. Finally, we hypothesize, with appropriate caution, that such nudges may be beneficial for students with attention deficit hyperactivity disorder, which affects a small but growing proportion of students (Green & Rabiner, 2012). Recent studies imply that consistent email reminders for assignments may be a useful time management strategy to support such students (Carroll et al., 2022).

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CONFLICT OF INTEREST STATEMENT

We report no potential conflicts of interest.

DATA AVAILABILITY STATEMENT

The data underlying this study are protected under the Family Educational Rights and Privacy Act (FERPA). Access to these data is contingent on approval from the University of Michigan per FERPA guidelines and regulations. Inquiries and requests for student data should be sent to the Center for Academic Innovation at cai-research@umich.edu.

ETHICS STATEMENT

This study was determined to be exempt from oversight by the University of Michigan's Institutional Review Board under application #HUM00119869. The study was conducted as

a secondary data analysis and under the FERPA exception for educational research given that the use of ECoach is now a standard part of the institution's educational practices.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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